Figure B.1 shows the workings of two independent pipelines with different sets of processors. These pipelines are hosted/created on an ingest node. The data is massaged while going through these processors before indexing. We can invoke these pipelines during a bulk load or when indexing individual documents.

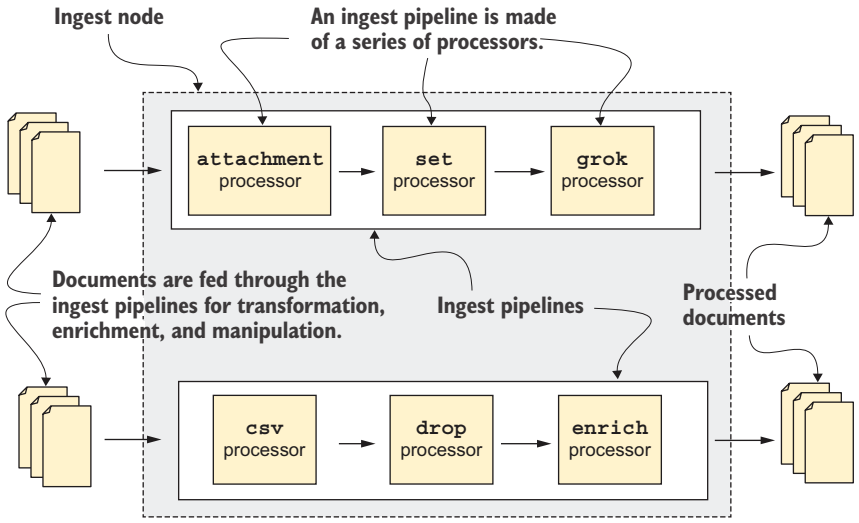


Figure B.1 Two independent pipelines processing data

A *processor* is a software component that performs one transformation activity on the incoming data. A *pipeline* is made of a series of processors. Each processor is dedicated to performing its one task. It processes the input based on its logic and spits out the processed data for the next stage. We can chain as many processors as our requirements dictate.

Elasticsearch provides over three dozen processors out of the box. Let's look at the mechanics of creating and using pipelines.

B.2 Mechanics of ingest pipelines

Ingest pipelines help us transform and manipulate data with minimal or no costly setup. A pipeline is made of a chain of processors, each performing a specific operation on the incoming data. These ingest processors are loaded onto nodes that are assigned an `ingest` role; remember from chapter 14 that each node in a cluster can be assigned a role—`master`, `data`, `ingest`, `ml`, and so on.

Let's take an example of MI5's top secret operations data being loaded from a database into Elasticsearch to integrate the search feature. The data extracted from the database must be stamped "confidential" using a `category` field (we can extend this example with further features later) before being indexed into Elasticsearch..